

100 Days of Azure DevOps

Day 41 Handout — If it is not in code, it is a rumor. Declarative desired state, idempotent apply, drift is the Portal's revenge

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — three ways to change Azure, only two are IaC

Style	What you write	Engine
Imperative script	az group create; az storage account create ...	You. Second run often duplicates or errors
Declarative ARM/Bicep	resources: desired JSON/Bicep	ARM control plane. Deployment history in Azure
Declarative Terraform	resource "azurerm_..."	Plan/apply + state file mapping address → Azure ID
Click-ops	Portal clicks	A rumor. No PR, no diff, no rollback of intent
Idempotency test	Apply twice	Same world. If you get two of everything, it was a script

Architecture B — pick one primary tool for Phase 5

Pick	When it fits a personal lab	Cost of picking both as primary
Bicep	Azure-native, no state file, what-if, modules	You will still read ARM errors — that is a feature
Terraform	State, locking, modules, travels off Azure later	You now operate terraform.tfstate like production data
ARM JSON only	Literacy (Days 42–43), not the mini project default	Eye strain without Bicep's compiler
Both as 'primary'	Never for this phase	Two half-labs. Pick one; skim the other

Architecture C — drift and other rumors

Pitfall	What you see	Fix
Portal snowflake	Firewall rule that exists only in prod	what-if / terraform plan before the next apply
Script of creates	Second apply: name already taken	Declarative resource with a unique name function
No decision record	Day 50 you cannot say why you picked the tool	docs/iac-why.md today
Treating drift as 'ops'	File green, Azure fork	Either import/correct the file or the click was an incident

One-liner to remember

If it is not in code, it is a rumor — IaC turns 'someone clicked prod' into a diff.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal repo azure-100-labs. No employer landing-zone stories. You will still touch the non-primary tool for literacy.

- 1 Write docs/iac-why.md: one drift you have actually seen (SKU bump, extra firewall rule, lock only in prod) — keep it generic, no client names.
- 2 Decision record in the same file: Tool: Bicep | Terraform. Reason: one paragraph. Not 'both'.
- 3 Add a two-line glossary: declarative = desired state; idempotent = apply twice, one world.
- 4 List the complementary tool as literacy only (ARM JSON if Bicep; Bicep sample if Terraform).
- 5 Do not deploy anything yet. Day 42 is broccoli (ARM). Day 44/46 is the primary track.
- 6 Commit the ADR on a personal branch. Invite nobody.

Decision record stub

```
# docs/iac-why.md
# Drift I have seen: _____
# Primary tool for Phase 5: Bicep | Terraform
# Reason: _____
# Literacy (not primary): ARM JSON always; the other DSL skimmed
# Idempotency test I will use: apply twice to rg-dayNN-lab, expect 0 extra resources
```

Done checklist

- Primary tool chosen with a reason, not a personality
- Drift example written without employer/client names
- Can define declarative vs imperative in one sentence each
- No Azure spend today

Tomorrow — Day 42

ARM templates basics — \$schema, apiVersion, az deployment group create. Literacy, not a love affair.